

The Book on Modal Triplet Theory

An Interpretive Guide

How to picture the program, what it has achieved, and what remains open

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Version 11

Abstract

A plain-language interpretive guide to Modal Triplet Theory. The book explains the program's central picture of one connected physical ontology described through compatible projections, then follows that picture through coherence, world-in-world geometry, the shared circle, q79/Fu–Yau geometry, proto-spinors, particles, the Standard Model reconstruction, quantum theory, gravity, time, memory, and cosmology. Short technical boundaries distinguish established results, declared reconstructions, and open interpretations, while proofs remain in their focused papers.

What This Book Is

This is the plain-language book about Modal Triplet Theory (MTT). It is meant to help the reader see the idea before meeting the machinery.

MTT has grown into a large research program. Its technical papers discuss Hilbert bundles, spectral projectors, fixed points, finite algebras, heterotic geometry, spin lifts, response matrices, and numerical certificates. Those objects matter, but they are not the best place to begin. The purpose of this book is to explain why they are being assembled and what kind of picture they may form together.

The guiding possibility is simple to state. The many theories used in physics may be effective descriptions of one richer structure. A description remains physical when it is stable, mutually compatible with the other descriptions, and capable of returning to a coherent state after disturbance. What we call particles, forces, spacetime, and quantum behavior may then be different visible faces of that common structure.

This is an interpretation of the MTT program, not a proof manual. The book owns no theorem and should not be cited in place of the focused papers. Technical boundaries are included where an attractive picture would otherwise outrun the results. They are deliberately short. Readers who want the proofs can follow the paper map near the end of the book.

In plain language. The book asks the imaginative question: what might the mathematics mean? The papers ask the stricter question: what exactly follows from the stated assumptions?

Version 11 Revision Note

Supersedes. *The Book on Modal Triplet Theory: An Interpretive Guide*, version 10.

Reason. Version 10 established the correct claim tiers and withdrawals, but its presentation still followed the technical corpus too closely in places. The book also needed a cleaner separation between its public description, its revision history, and the papers that own the mathematical results.

Resolution. Version 11 restores the book’s intended role as an interpretive guide. It develops the central pictures in ordinary language, keeps formulas subordinate to explanation, attributes technical claims to their paper owners, and keeps this revision record outside the descriptive synopsis used for publication metadata.

Retained content. The claim-typed overview, corrected q79 and shared-circle interpretation, Standard Model profile status, and explicit withdrawals of obsolete claims are retained without theorem promotion.

Remaining boundary. The book remains an interpretive map and proves no independent theorem. The open source, action, quantum-probability, continuum-QFT, gravity, and cosmology bridges remain exactly those recorded by the current technical papers and A/B ledger.

Version 10 Revision Note

Supersedes. *The Book on Modal Triplet Theory*, version 9 (January 2026).

Reason. The original edition succeeded as an intuitive account, but it sometimes described promising interpretations as if the technical papers had already proved them. Later work also replaced the literal circle–lens–nil topology, corrected the meaning of the shared circle, selected a q79/Fu–Yau branch, clarified the local 3-by-3 comparison geometry, and completed a much stronger Standard Model reconstruction.

Resolution. This edition restores the original book’s explanatory purpose while bringing its claims into line with the current papers. The mathematics is mostly translated into pictures and ordinary language. Statements that remain proposals are presented as proposals, and proved results are attributed to their technical owners.

Retained result. The central interpretation survives: stable physical descriptions may arise through a common geometry of comparison, projection, return, and admissibility. Several pieces of that picture now have exact or reproducible mathematical realizations.

Remaining boundary. MTT has not yet derived the observed universe from a unique parameter-free microscopic source. The decisive missing bridges include the global q79 spin decision, the local-to-global geometry map, a single selected action and value source, an MTT derivation of quantum probability, and a controlled derivation of gravity and cosmology.

Version 9 is preserved as the historical edition. The changes above do not erase its ideas; they separate the ideas that remain fruitful from claims that were too strong. In particular, this edition does not identify a compact phase circle with physical time, does not treat a literal S^1 –Lens–Nil product as the selected universe, does not derive Born probabilities by counting basins, and does not claim completed quantum gravity or cosmology.

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Part I

The Guiding Picture

Chapter 1

Why Look for Another Language?

Physics is extraordinarily successful and strangely divided. General relativity describes gravitation through spacetime geometry. Quantum theory describes states, amplitudes, and measurements. Quantum field theory turns particles into excitations of fields. The Standard Model organizes almost everything observed in particle laboratories, yet leaves many measured numbers as inputs. String theory, spectral geometry, statistical mechanics, and dynamical systems each reveal another aspect of the same world.

This division may reflect nature, but it may also reflect our choice of language. A map of a city, a railway timetable, and a geological survey can all be correct while looking nothing alike. Their symbols are different because their questions are different. Trouble begins when one mistakes a map for the city itself or assumes that two maps must be unrelated because their symbols do not match.

MTT begins from the suspicion that some divisions in physics are divisions between maps. It asks whether a richer reference system can support several familiar descriptions at once and explain why each becomes accurate after the appropriate projection. The ambition is not merely to put existing theories side by side. It is to understand the transformations that turn one level of description into another.

This changes the kind of question being asked. Instead of starting with a list of particles and forces, one asks what allows any persistent physical description to exist. Instead of treating a parameter as an unexplained dial, one asks whether it is the visible shadow of a relation in the upper geometry. Instead of checking complicated rules only after projection, one looks for a simpler rule before projection from which those rules descend.

That last possibility is especially important. Gauge symmetries, particle labels, and quantum rules may appear complicated because they are expressed in a reduced language. The same information may be simpler in the larger space, just as a rotating three-dimensional object can cast a complicated two-dimensional shadow.

Technical boundary. No theorem presently shows that all established theories are projections of one MTT object. The current program contains exact modules, conditional bridges, and a broad Standard Model reconstruction. A universal preprojection language remains a research target.

Chapter 2

One World, Several Descriptions

The original intuition behind MTT was ontological economy: reality should not need a separate fundamental substance for every way we describe it. The electron, its wavefunction, the electromagnetic field around it, the curvature it contributes to, and the record left in an apparatus should not be five unrelated worlds. They should be aspects of one physical process.

The word *modal* is useful here. A mode is a way in which a structure can respond, persist, or vary. The same instrument can have many normal modes; the same geometric carrier can support many effective fields. MTT imagines a richer carrier whose stable modes appear, after reduction, as the familiar objects of physics.

This does not mean that every description is equally fundamental or that all are interchangeable everywhere. A fluid description is excellent for an ocean and poor for an individual molecule. A quantum field description can be indispensable at one scale while a geometric description is more natural at another. MTT therefore speaks of domains, projections, and admissibility. A description is used where its map is controlled.

The phrase “one field” in this book should be read in this broad, interpretive sense: one connected physical ontology rather than a proven single scalar or spinor field with a final equation. The present technical program is more general. It works with bundles of state spaces and compatible operators, allowing different local descriptions to be carried together without pretending that they are already identical.

This is a useful correction to the first edition. The unity sought by MTT is not uniformity. A single language can contain distinctions. Indeed, it must. Quarks and leptons behave differently; bosons and fermions transform differently; time and phase are not the same; local strain and global topology are not the same. The aim is to make these distinctions arise in one coherent architecture rather than remove them by verbal analogy.

In plain language. MTT does not ask every theory to say the same thing. It asks whether their different statements can be traced to compatible views of one underlying structure.

Chapter 3

Projection Is Not Destruction

Projection is one of the central images of MTT. In ordinary life, projection often sounds like loss: a three-dimensional object casts a flatter shadow. In mathematics and physics, however, projection can also mean selecting the part of a state relevant to a question.

A camera does not destroy the object it photographs. It records particular relations between the object, the lens, and the detector. A musical score does not contain the air pressure at every point in a concert hall, yet it retains the structure needed to reproduce the music. A thermodynamic state does not list every molecular coordinate, but it can still make exact macroscopic predictions.

MTT treats the visible world in a similar way. A projected description may omit upper information while preserving the combinations that remain stable and observable in its domain. Different projections may retain different aspects. Their compatibility is not automatic; it is something the theory must check.

This is why projection alone cannot explain everything. If two different upper states produce the same visible state, a reduced observer cannot infer which upper state occurred without more information. A right section may choose a representative, but choosing one is not the same as recovering the actual one. Likewise, a many-to-one map does not generate probability, entropy, or quantum uncertainty by itself. Those require states, measures, and dynamics.

The deeper hope is constructive. Perhaps the rules that look arbitrary after projection are consistency conditions among upper descriptions. A gauge rule might express the freedom to choose a local frame. A conservation law might record an invariant that survives reduction. A family label might record where a mode sits in a recursive carrier. If so, the right preprojection language would turn several separate postprojection rules into one geometric fact.

Technical boundary. The Projection–Admissibility paper gives exact criteria for when an upper evolution descends to an autonomous reduced evolution and when information can be recovered. It does not claim that every physical projection satisfies those criteria.

Chapter 4

Coherence, Return, and Admissibility

Projection says which part of a description is retained. Coherence says whether the retained parts belong together. Return says whether a disturbed state can settle back into a stable regime. Admissibility says whether the whole route remains within the domain where the description is valid.

These ideas are familiar outside MTT. A bridge is coherent when its parts carry load together. A laser is coherent when phases remain related. A biological organism maintains itself by repeatedly returning toward viable states. A numerical method is admissible only while its approximations remain controlled.

MTT combines these intuitions but does not collapse them into one word. A state may be spectrally coherent without being a dynamical equilibrium. A projected step may have a fixed point without the upper system being at rest. A stable local solution may fail when continued around a nontrivial global loop. A calculation may reproduce a measured profile while leaving the source of that profile unexplained.

The early slogan of one inequality selecting one inevitable world was therefore too compressed. The current program uses several gates. The state space must be well defined. The relevant operators must be compatible. The evolution must exist. The selected domain must remain invariant. Disturbances outside the coherent sector must be controlled. If uniqueness is claimed, a valid uniqueness mechanism is needed. If equilibrium is claimed, return under one projected step is not enough.

This may sound less dramatic than the original slogan, but it makes the idea more powerful. Each gate can be tested. A failure has a location. Different physical encodings can share some gates while differing at others. This is how an interpretive picture becomes a research program.

In plain language. Coherence is not a magical force. It is the demand that several descriptions remain mutually compatible and dynamically stable on a stated domain.

Part II

The Geometry Behind the Picture

Chapter 5

The Upper World and the Visible World

The canonical physical picture used by MTT has four extended spacetime directions and a six-dimensional internal carrier. This is the familiar ten-dimensional shape encountered in several unification programs, but MTT uses it in its own way.

The four-dimensional side is the world in which clocks, causal signals, and laboratory positions are described. The six-dimensional side carries internal comparisons, phases, representation data, and modes that need not look like extra corridors in ordinary space. It is better to picture the internal directions as ways the physical structure can be arranged than as tiny places into which a particle could travel with a sufficiently small ruler.

The distinction matters because a dimension can play different roles. Spacetime dimensions locate events. Internal dimensions can label orientations, phases, shapes, or fibers over spacetime. A six-dimensional fiber attached to every spacetime point is not another six-dimensional universe sitting beside ours. It is additional structure carried at each point of the visible description.

MTT does not currently prove that nature must have this four-plus-six form. It selects and studies that realization because it can host the geometric, spinorial, heterotic, and finite-operator structures used by the program. The abstract fixed-point mathematics itself is dimension-neutral.

The internal and external sides are not meant to be sealed compartments. Their interaction is the point. Internal patterns determine what can appear as a field or particle in four dimensions, while visible dynamics can disturb the internal configuration. The eventual theory must explain this exchange through one action and one controlled reduction.

Current status. The four-plus-six realization is a supplied physical arena with strong supporting constructions. Its unique selection from more primitive MTT data is still open.

Chapter 6

The World-in-World Comparison

One of the most suggestive MTT ideas is that a world can carry a structured comparison with another world-like frame. The phrase “world in world” does not mean a smaller universe hidden inside a larger one. It means that two three-directional structures can be compared point by point.

Imagine holding two small coordinate frames. To say how one sits relative to the other, one needs nine comparison numbers: each of three directions in the first frame can be compared with each of three directions in the second. That is the source of the 3-by-3 object. It is a field of comparisons, not a multiplication of manifold dimensions.

Those nine comparison directions divide naturally. Three describe relative rotation. The remaining six describe strain: one overall expansion or contraction, two diagonal shape changes that preserve total size, and three off-diagonal shears. If one also supplies a single ordering direction, the component count can be summarized by the book’s one useful mnemonic:

$$1 + 3 \times 3 = (1 + 3) + (1 + 2 + 3) = 4 + 6 = 10.$$

The identity is exact as a count of components. Its interpretation is the interesting part. The ordering direction and three rotations suggest an external four-dimensional lane. The six strains suggest an internal lane. This is a possible bridge between the original world-in-world intuition and the ten-dimensional carrier used elsewhere in MTT.

But a component count is not yet a universe. To become physics, the comparison field must be attached globally, supplied with a metric and connection, and related to the operators that determine particles and responses. The current work has made the local decomposition precise. The global identification is one of the main frontiers.

In plain language. The 3-by-3 field tells us how two three-directional frames differ. Rotation uses three directions; deformation uses six. It is a comparison language, not nine ordinary spatial dimensions.

Chapter 7

Why the Pattern One, Two, Three Appears

The title Modal Triplet Theory can tempt the reader to treat the number three as a mystical premise. That would be a mistake. The current program does not begin by declaring that everything comes in threes.

What does recur is a rank pattern: first one direction is active, then two, then three. A flag in a three-dimensional space works this way. One begins with a line, extends it to a plane, and then reaches the full space. The new content added at each stage has sizes one, one, and one, while the cumulative carriers have ranks one, two, and three. In the strain picture, a related decomposition has one scalar direction, two diagonal shape directions, and three shear directions.

This is *recursive activation*, not three independent copies of the same space. The first lane is intentionally smaller. The second reuses what came before and adds structure. The third reuses both and completes the carrier. That reuse is one reason the original language of nesting felt natural.

There is also an important symmetry warning. Before a flag is chosen, the five shape-changing strains form one rotationally natural block. The more detailed split into two and three requires a selected frame or flag. MTT must therefore explain where that selection comes from; it cannot treat the decomposition as simultaneously frame-free and fully resolved.

The triplet is best understood as an organizational principle that appears in several successful constructions. It may label operator lanes, bundle summands, representation sectors, or stages of a filtration. Whether one common source ultimately selects all of these appearances remains an open and testable question.

Chapter 8

Circle, Lens, and Nil

The original MTT vocabulary used three memorable geometric images: circle, lens, and nil. They remain useful, but their meaning has become more precise. They should not be pictured as three literal manifolds nested like shells.

The *circle* represents shared phase or holonomy. A phase can return to its starting value after a complete turn while still retaining information about how it travelled. In bundle language, the same circle-like degree of freedom can be reused across several sectors. It is counted once, not once per sector.

The *lens* represents finite or projective transport. A lens space is a concrete geometric model with nontrivial global identifications, but the wider MTT role is the controlled distinction between paths or sheets that can look the same locally. The binary phase choice often described as $+i$ or $-i$ belongs naturally to this projective or signed transport intuition.

The *nil* represents triangular transport, shear, anchoring, and termination. Nilpotent structures remember ordered influence: one direction can feed another without every influence being reversible in the same basis. The three-dimensional Heisenberg nil group is a standard mathematical model for such triangular behavior.

Together, circle, lens, and nil provide a language for recurrence, finite identification, and ordered deformation. This language can organize an operator filtration even when the global internal manifold is something else.

The shared circle is not physical time. Time in the physical completion is a noncompact causal order. A circle may synchronize phases along that order, and a line can cover a circle repeatedly, but that relation requires a map and a frequency. A clock hand is not time itself.

The idea of a double traversal remains an evocative interpretation. One traversal may carry a state through a visible branch while a second completes the phase or closure relation. Perfect return could then correspond to zero residual strain, whereas an incomplete visible return could appear as memory, time direction, or gravity. At present this is a hypothesis to be modeled, not a result used in the proofs.

Technical boundary. The literal products $S^1 \times \text{Lens} \times \text{Nil}$ and $L(3, 1) \times \text{Nil}_3$ are not the selected q79/Fu-Yau compactification. They remain auxiliary models and sources of operator intuition.

Chapter 9

The q79 Global Candidate

Local pictures are easy to draw and hard to globalize. A coordinate frame can always be chosen in a small neighborhood. The real test is what happens when the frame is carried around a nontrivial loop and returned to its starting point.

The q79 branch is the most developed MTT attempt to provide that global carrier. It uses a degree-three spectral cover in a Fu–Yau/Hull–Strominger setting. The details belong to algebraic geometry and heterotic compactification, but the intuitive point is accessible.

A three-sheeted cover contains three levels of information. There is the part common to all sheets, the two-dimensional variation whose total trace vanishes, and the full three-dimensional sheet algebra. This produces another one-two-three profile. Unlike the local strain split, it is built from the trace structure of a global complex carrier.

The branch label $q = 79$ is not a fitted particle mass or a mysterious new constant of nature. It is selected by an exact finite arithmetic construction inside the declared branch and quotient. That is a real achievement, but it does not make every physical value follow from 79.

The q79 geometry also carries local spin information. Signed permutations of the three sheets can be lifted to a double cover, and the local relations can be checked exactly. The remaining global question is whether every relevant loop returns with the correct central sign. If not, the natural result may be a SpinC structure in which the shared circle participates in the cancellation rather than a strict Spin structure.

Current status. The q79/Fu–Yau branch is the strongest selected global compactification candidate in the current corpus. Its strict global spin obstruction is not yet fully decided.

Chapter 10

The Bridge We Still Need

The local world-in-world carrier and the global q79 carrier both display a one-two-three pattern. This resemblance is why they are exciting together. It is also where caution matters most.

Two rooms can have the same dimensions without being the same room. Two vector spaces can have the same rank without carrying the same metric, connection, symmetry, or dynamics. The local strain carrier is real and geometric. The q79 trace carrier is complex and algebro-geometric. Their matching rank is an invitation to construct a map, not permission to assume one.

The missing bridge must do more than pair basis vectors. It must work from patch to patch, respect the selected flag, preserve inner products and connections, and carry the relevant response operators and overlap data into one another. In physical language, it must show that the local way the medium can strain is the same structure that the global compactification uses to organize its modes.

If this bridge exists, several strands of MTT would become one chain. The world-in-world picture would supply an intuitive local origin for the six internal directions. The q79 geometry would tell those directions how to fit together globally. The finite operators used in the Standard Model reconstruction could then be tested as reductions of the same source rather than as a compatible but separately selected architecture.

If no such bridge exists, the rank resemblance remains a useful analogy but not the foundation of one theory. Either outcome would be informative. This is why the bridge is a better research target than another numerical coincidence.

Part III

From Geometry to Particles and Forces

Chapter 11

The Proto-Spinor

A vector rotated through a full turn returns to its original direction. A spinor carries subtler memory: a full turn need not return its sign, while a double turn does. This familiar double-cover behavior is one of the clues behind the MTT proto-spinor.

The starting point is orientation. A three-directional frame can be moved around a loop that cannot be shrunk away inside the rotation group. An ordinary vector description forgets this loop class. Passing to the double-covering spin group retains it. In that precise sense, spinorial behavior is the minimal way to remember a certain kind of rotational history.

MTT uses the term *proto-spinor* because this is not yet a physical electron or quark. It is preprojection orientation data from which a spinorial chart may be built. To become a physical fermion, the construction still needs a Lorentzian spacetime spin or SpinC structure, a tetrad, connection, Dirac operator, statistics, states, and observables.

The distinction is valuable. Conventional particle theory begins with spinors as the right representations of spacetime symmetry. MTT asks whether the need for those representations can be understood as the visible encoding of loop memory in a richer carrier. It does not replace the conventional description; it looks for its geometric source.

Weyl, Dirac, Majorana, Clifford, and twistor languages can then be viewed as different ways of organizing the completed spinorial data. Clifford algebra records how directions multiply and how the metric is represented. Weyl language isolates chirality. Dirac language pairs chiral sectors. Twistor language reorganizes conformal and causal geometry. The proto-spinor is not a rival to these constructions. It is a proposed common preprojection carrier from which the appropriate chart might descend.

In plain language. The proto-spinor is a memory-bearing orientation object. It explains why a double cover is natural; further physical structure is required before it becomes a fermion.

Chapter 12

Particles as Persistent Patterns

The interpretive heart of MTT is that a particle need not be a tiny indestructible bead. It can be a persistent pattern of the common carrier: a mode that survives projection, responds consistently to disturbances, and returns to its admissible sector.

This way of speaking is not foreign to physics. A photon is already understood as an excitation of a field. A quasiparticle in a material is a stable collective mode rather than a new microscopic substance. A soliton is a pattern protected by dynamics and topology. MTT extends that style of thinking to the full particle inventory.

Different particles would correspond to different ways a mode occupies the internal carrier. Charge would record how it transforms under phase and gauge transport. Spin would record its response to frame rotation and the double-cover structure. Color and weak labels would record how the mode sits in the selected finite representation. Mass would describe the energetic or dynamical cost of exciting the normalized mode, not merely the curvature of an arbitrary unnormalized function.

The original book associated each force or family directly with one of three geometric bundles. That assignment was too literal. The current picture is more layered. Circle, lens, and nil provide carrier roles. The q79 geometry organizes global modes. A finite algebra organizes the observed gauge representation. A finite Dirac operator organizes couplings and masses at the reconstruction tier. These layers must be connected by source maps; one cannot infer the particle table from the number three alone.

There remains an intriguing possibility that quarks represent an additional order of internal breakdown or closure strain beyond leptons. Quarks carry color, are confined, and participate in a richer mixing and threshold structure. A second-order response or a deeper nil/shear activation could provide a geometric distinction. This is a promising model-building clue, not a current derivation.

Technical boundary. The finite-operator work shows that the chosen MTT architecture can carry the Standard Model particle representations and profiles. It does not yet prove that those representations are the unique particles selected by the upper geometry.

Chapter 13

Forces as Rules of Comparison

A force can be described as a push, but modern physics often describes it more deeply as geometry. Electromagnetism is organized by a phase connection. The weak and strong interactions are organized by non-Abelian gauge connections. Gravity is organized by spacetime geometry.

MTT interprets these structures as rules for comparing neighboring descriptions. A connection tells us how to carry an internal frame from one point to another. Curvature tells us whether the frame returns unchanged around a loop. Gauge freedom says that different local frame choices can describe the same physical relation.

From this point of view, the gauge groups seen after projection may be the symmetries of a selected comparison structure. The upper world need not carry the final laboratory labels in their familiar form. It must carry enough geometry that the correct labels and transformations descend.

The current finite branch uses the familiar algebraic ingredients associated with the Standard Model: complex phase, quaternionic weak structure, and three-color matrix structure. On the selected particle carrier, anomaly conditions recover the usual hypercharge direction and global gauge quotient. This is not produced by the bare one-two-three strain count. It is a result of the selected finite representation.

The long-term MTT goal is to reverse the logic. Instead of supplying the finite algebra and checking that the right gauge rules follow, one would derive that algebra from the shared q79 and comparison geometry. Then the complicated postprojection gauge rules would become the natural shadow of a simpler upper transport law.

Chapter 14

What the 27 by 27 Architecture Means

The number 27 can sound mysterious, especially beside the many other dimensions used in particle physics. It should be understood as the size of a particular finite response architecture, not as a new number of spacetime dimensions and not as a universal matrix size in quantum mechanics.

Quantum mechanics can use matrices of any size appropriate to a chosen state space. A two-level system uses two-by-two matrices. A three-level system uses three-by-three matrices. Composite systems use tensor products, so their dimensions multiply. The MTT 27-by-27 construction similarly comes from a chosen organization of three-level and sector data. Its value lies in how the blocks are arranged and connected, not in 27 by itself.

The matrix acts as a compact bookkeeping surface for several kinds of response. Some blocks organize gauge and family structure. Some carry orientation and phase. Some connect Higgs, Yukawa, and threshold information. Zeros are as important as nonzero entries because they state which couplings are forbidden or absent at that tier.

A larger 96-dimensional finite Hilbert space and its 96-by-96 finite Dirac operator provide the particle–antiparticle and three-family realization used for the full profile check. The 27-by-27 and 96-by-96 objects therefore serve different but related purposes. One is a minimal response architecture; the other is the fuller finite particle carrier.

What has been achieved is not that a bare integer matrix predicts nature. The achievement is that one finite architecture can consistently host the gauge representation, family organization, flavor mixing, phase, Higgs, threshold, and precision data needed for a broad Standard Model reconstruction. The remaining question is whether the entries themselves are emitted by the same selected upper geometry.

In plain language. The matrix is a structured finite model of how sectors talk to one another. Its shape is meaningful because of its blocks and constraints, not because all of physics normally uses 27-by-27 matrices.

Chapter 15

Masses, Mixing, and the Standard Model Result

Masses and mixing angles are where many unification ideas become vague. It is easy to reproduce a symmetry. It is much harder to reproduce the numerical pattern of electron, muon, tau, quark, Higgs, and threshold data while keeping track of what was fitted.

The current MTT calculation program has achieved a broad result at a clearly defined standard. The finite matrix, charged Yukawa profiles, CKM mixing, electroweak rows, Higgs and threshold responses, finite Dirac operator, multi-loop transport, and precision covariance have been joined into one consistent renormalized Standard Model reconstruction. The consolidated ledger closes all twelve obligations at the adopted one-shared-physical-primitive/profile tier.

This means the architecture can represent the measured Standard Model and transport its data consistently using one shared physical normalization and a common scheme. It is stronger than a collection of unrelated curve fits. The sectors are required to agree inside one frozen finite and renormalized construction.

It is not yet the strongest possible result. Measured renormalized profile values are admitted. Standard quantum-field-theory quantization is imported. The one shared physical primitive supplies a normalization rather than being derived uniquely from the upper geometry. The result is therefore a reconstruction or profile equivalence, not a parameter-free prediction of all masses and mixings.

This distinction does not make the reconstruction empty. The ordinary Standard Model also takes its Yukawa values and several other quantities from experiment. MTT has shown that its selected architecture can carry the whole profile coherently and with fewer independent structural choices at its declared tier. To surpass the Standard Model as an explanation of those numbers, it must now derive the value rows from a source fixed before looking at them.

The most credible route is not to keep fitting ever more elaborate formulas. It is to compute normalized overlap kernels and response operators from one selected q79/Fu-Yau action and test their outputs against data that were not used to choose the source.

Chapter 16

Theta, Thresholds, and a Useful Failure

The Theta program was created to connect geometric and finite data to renormalized physical quantities. It has produced important transport and consistency machinery. It also produced one of the program's most useful corrections.

An earlier chain appeared to select a crossing in the few-TeV range. For a time this looked like a possible new physical scale. Later audits showed that the chain mixed calibrated gauge information, auxiliary geometry, and round-trip identities in a way that did not support the claimed prediction. The old few-TeV chain is withdrawn, and the crossing was retired.

This was progress. A framework that cannot relinquish an attractive number is not doing research. The corrected Theta program now treats its gauge ratios as measured profile coordinates transported to a common scale. The weak-angle relation obtained from the same ratio is a consistency check, not an independent prediction. Auxiliary round-sphere and nil models remain cross-checks rather than the selected q79 volume.

The positive legacy is a sharper standard for future work. A source must be frozen before the target value is inspected. Inputs used for geometry, normalization, branch choice, and scheme choice must be listed. A prediction must be held out from all of them. This discipline now informs the broader Standard Model and precision program.

Part IV

Quantum Theory as a Visible Language

Chapter 17

Before and After Projection

Quantum mechanics is usually presented in its own native language: complex Hilbert spaces, operators, amplitudes, unitary evolution, and the Born rule. MTT does not deny this language. It asks whether the language describes the visible side of a richer construction.

Before projection, the proposed MTT reference system contains geometric carriers, shared phase, orientation memory, spectral modes, and admissible flows. After projection, some combinations may be represented as quantum states and observables. The hope is that rules that look postulated in the quantum description will become compatibility conditions of the upper structure.

There are suggestive correspondences. A spectral subspace resembles a space of allowed states. A projector resembles a quantum alternative. A connection supplies phase transport. A double cover supplies spinorial behavior. Noncommuting effective observables might arise from incompatible reductions. Stable sectors resemble superselection sectors.

Correspondence is not derivation. To recover quantum mechanics, MTT must produce the complex Hilbert structure, observables, composition rules, dynamics, and probability functional with the correct operational meaning. The present mathematical-language program is searching for precisely this preprojection object: a common carrier from which Dirac, Weyl, Clifford, twistor, quantum, field-theoretic, and geometric charts descend without being inserted independently.

The advantage of looking before projection is conceptual economy. An awkward rule in one chart may be a simple relation in the common carrier. The danger is equally clear: one can always rename familiar quantum objects as projections without explaining anything. The test is whether the upper language forces relations that were independent below and makes new, held-out predictions.

Chapter 18

Measurement Is an Ordinary Physical Process

Measurement is often given a starring role in discussions of quantum theory. In the interpretation pursued here, it should not be privileged. A measurement is a physical interaction that leaves a usable record. It is not a special act by consciousness and not a metaphysical boundary where nature changes its laws.

An atom interacting with a detector, a photon scattering from dust, and a laboratory apparatus amplifying a microscopic difference are all physical processes. Some are called measurements because an observer later uses the record. The underlying event does not know that it belongs in an experiment.

MTT can describe such an event as a disturbance followed by physical continuation into a stable record sector. The apparatus, environment, and system are parts of the same carrier. Projection can explain why an effective description loses access to some upper distinctions. Fixed-point and admissibility ideas can help explain why records become robust.

None of this supplies probabilities automatically. The number or volume of stable basins depends on how states are measured and weighted. The Born rule requires a specific state functional and measurement instruments, or a theorem showing that the MTT weights equal quantum trace probabilities in the full relevant class of experiments.

Nor can a deterministic upper story simply ignore Bell's theorem. If MTT eventually provides a deeper deterministic account, it must say exactly where Bell factorization fails: perhaps through contextuality, global boundary consistency, nonseparability, or another explicit mechanism. It must preserve the observed no-signaling structure.

In plain language. Measurement is not the source of reality. It is one ordinary way physical systems interact and create records. The unresolved question is why the statistics of those records obey the Born rule.

Chapter 19

Quantum Field Theory and Locality

Quantum field theory combines quantum rules with locality, relativity, gauge symmetry, and renormalization. Reconstructing its particle data is not the same as deriving the theory.

At the current Standard Model tier, MTT supplies a finite representation and response architecture, then uses standard BRST/Faddeev–Popov quantization and multi-loop transport. This is the correct procedure for testing profile equivalence. It also marks the boundary: the path integral, gauge complex, and interacting continuum theory are not yet consequences of projection.

Locality is a particularly important test. A reduction performed separately over each spacetime point can preserve the locality of an upper local theory when the selected sector is invariant. By contrast, an instantaneous kernel that reaches across space does not become causal merely because its leading differential operator looks hyperbolic. A local mediator or a genuinely causal retarded kernel is needed.

This is where MTT’s insistence on typed maps helps. A relation between distant records may be global in the upper state without permitting a controllable signal to propagate faster than light. But that sentence is only an interpretation until a state space, local algebra, dynamics, and no-signaling proof are supplied.

The long-term goal is a common source from which the finite particle architecture and the continuum quantum field theory descend together. That would turn present profile equivalence into a deeper physical equivalence.

Chapter 20

Strings, Worldsheets, and Fu–Yau Geometry

String theory provides one of the richest existing languages for connecting geometry, quantum fields, gauge structure, and extra dimensions. MTT has much to learn from it. The relationship is not that MTT has already contained or replaced string theory.

The proto-spinor and worldsheet constructions identify a local bridge between two descriptions near an aligned configuration. When their quadratic responses are intertwined, they agree to controlled local order. This gives a mathematical reason why a two-dimensional worldsheet language may encode some proto-spinor dynamics.

A complete string theory requires much more: Weyl and modular consistency, ghosts, spin structures, anomaly cancellation, beta functions, unitarity, and target-space matching. Those obligations do not disappear because two local Hessians agree.

The strongest established meeting point is Fu–Yau/Hull–Strominger geometry. It supplies genuine heterotic flux compactifications on non-Kähler spaces. The q79 branch lives naturally in this setting. MTT can therefore use mature string-geometric tools while asking a different selection question: why this carrier, this cover, and these finite responses?

Fixed-point methods may also be useful to string theory in their own right. They provide a disciplined way to ask whether a chosen compactification, bundle, or flow returns to a stable admissible sector and whether a finite truncation remains controlled. This is a potential contribution even before MTT becomes a complete physical theory.

Part V

Gravity, Time, and the Larger Story

Chapter 21

Gravity as Closure Strain

The original book pictured gravity as the elastic response of the common medium. That remains one of MTT’s most intuitive ideas, provided it is stated carefully.

Imagine that the upper carrier has preferred relations among its internal directions. Matter and quantum fields disturb those relations. The carrier responds by changing the geometry through which visible processes propagate. From the projected four-dimensional viewpoint, that response can look like curved spacetime.

The world-in-world strain field gives this analogy real mathematical content. It separates rotation from deformation and identifies scalar, shape, and shear responses. The closure Hessian measures the local cost of such deformations on a declared slice. The q79 and finite-operator work provides candidate internal sources and modes.

But elasticity is not yet Einstein gravity. A physical theory must supply a Lorentzian metric, gravitational action or equations, constraints, stress-energy coupling, causal initial-value problem, and normalization of Newton’s constant. It must show that unwanted internal modes do not spoil the four-dimensional equations or quantify the error when they are eliminated.

The current ten-dimensional action paper is therefore an ansatz on a stated regime, not a final derivation. It includes familiar gravitational, gauge, matter, and flux ingredients so that the reduction can be studied. The next step is to select that action from the common geometry rather than choose it because it has the desired low-energy form.

In plain language. MTT’s picture is not “gravity is a new force in an elastic substance.” It is that the geometry seen as gravity may be the visible response of a richer constrained carrier to physical stress.

Chapter 22

What This Means for Quantum Gravity

The phrase “quantum gravity” can suggest that one must take the classical gravitational field and quantize it in the same way as another field. MTT points toward a different emphasis. Quantum fields and quantized actions may already be modes of the common carrier. Their activity places stress on that carrier, and the geometric response is what the lower description calls gravity.

In this picture one is not ignoring quantum gravity. One is relocating the problem. The key object is the joint quantum-geometric response of the upper system, not a classical metric treated in isolation. A graviton could still appear as a small coherent excitation of the effective geometry, while the fundamental ontology is the larger response system.

This approach offers hope because it may avoid forcing incompatible descriptions together at the wrong level. The ultraviolet behavior of a projected field can be softened by internal spectra, finite projections, and nonlocal form factors that arise from a local higher structure. The q79 geometry and finite spectral algebra provide concrete tools for exploring that possibility.

The Planck scale remains essential. It marks the regime where the effective separation between matter on spacetime and the spacetime response can no longer be trusted. In MTT language, it may be the scale at which internal and external response channels can no longer be reduced independently. This is an interpretation, not a derived numerical Planck scale.

Filtered propagators can make selected integrals finite. That alone does not establish a UV-complete theory. Unitarity, causality, gauge consistency, constructive existence, and physical observables must all survive. The current QG work has useful conditional models and exact support packets, but the older claim of completed all-loop finite and unitary quantum gravity is withdrawn.

Chapter 23

Time, Phase, and the Direction of a Branch

Time is easy to confuse with any ordered parameter. A smoothing procedure has an order. Renormalization has a scale order. A circle has an angular order. A proof can proceed step by step. None of these is automatically physical time.

In the current MTT completion, physical time belongs to the noncompact Lorentzian spacetime description. It is the parameter along which causal evolution is posed. The shared circle carries phase or holonomy. A physical clock can link elapsed time to repeated phase, but the phase cycle and time remain different objects.

The branch idea adds another layer. The underlying equations or global carrier may admit more than one orientation or completion. Our universe may occupy one branch selected by initial or boundary data. The existence of an opposite mathematical branch does not require it to be physically realized, stable, or well behaved. It is similar to an equation allowing time-reversed solutions while the observed universe has a thermodynamic arrow.

The binary $+i/-i$ choice and the double-cover structure make this question especially natural in MTT. Perhaps both orientations belong to the full carrier while one is visible after branch selection. Perhaps a double traversal is needed for complete closure, with one traversal corresponding to the experienced branch. Or perhaps a global obstruction selects one sign. These are distinct possibilities and should not be merged before the relator and source theorems decide among them.

An arrow of time also needs more than a branch label. It requires an asymmetric state, law, or boundary condition. A stabilization function can orient a mathematical flow, but physical entropy requires a state, measure, and coarse-graining. The two may eventually be connected by the common carrier.

Chapter 24

Memory, Entropy, and Cosmology

The phrase “the universe has a bad memory” captures an important interpretive theme. A projected world may retain only part of its upper history. Many distinct microscopic routes can merge into one stable visible record. The universe can be lawful without carrying a readable copy of every detail in every effective description.

This limited memory may help explain why entropy and irreversibility appear. As correlations spread into degrees of freedom unavailable to a chosen projection, reconstructing the earlier visible state becomes increasingly difficult. Yet information loss in an effective description is not fundamental destruction. Whether the upper dynamics is reversible, how states are weighted, and what information remains globally are separate questions.

The original book extended this picture into a grand cosmology: a Great Coherence replacing the Big Bang, inflation as relaxation, dark matter as modal solitons, baryogenesis from geometry, and renewal after heat death. These remain imaginative scenarios, not current results.

A credible MTT cosmology must begin with the selected action and a valid cosmological solution. It must identify initial or boundary data, derive the background expansion, calculate perturbations, transport them to observable spectra, and compare them with data without using those data to select the same branch.

The interpretive picture is still useful. The early universe would be a regime in which the separation between internal and visible descriptions was changing rapidly. Inflation could correspond to a controlled relaxation only if the resulting perturbations and reheating follow. Dark sectors could be modes that interact gravitationally while remaining outside the ordinary finite particle carrier. These are questions the eventual action can answer.

Part VI

What the Mathematics Has Actually Secured

Chapter 25

Fixed Points Without the Formalism

A fixed point is a state that returns to itself under a stated operation. This simple idea appears throughout MTT because stable physical descriptions must survive repeated comparison and reduction.

There are several different notions hiding behind the phrase. A projected time step may return a state after one cycle. A continuous flow may have an equilibrium that never moves. A statistical system may have an invariant distribution even though individual states fluctuate. A curved family of spectral subspaces may track a moving center rather than one fixed point.

The six Fixed Points papers separate these cases. The first establishes the analytic setting, well-posedness, projectors, existence and uniqueness routes, and finite approximation. The second studies the supplied four-plus-six realization. The third treats deterministic disturbances, noise, and damping. The fourth follows curved spectral clusters and leakage. The fifth studies covariance, correlation, and exit risk. The sixth explains how those results fit together and what extra assumptions are needed for physical interpretation.

The important lesson is not that everything in nature is literally a fixed point. It is that claims of stability can be decomposed into checkable questions. Does the evolution exist? Does it remain in the admissible domain? Does some state return? Is it unique? Is it an equilibrium or only a stroboscopic return? How large can disturbances become before the reduction fails?

This framework is one of MTT's most mature contributions. It can be used in other settings, including compactification and effective-field-theory problems, without accepting every physical interpretation in this book.

Chapter 26

The Strongest Results in Plain Language

Several achievements now stand independently of the grand interpretation.

A reusable analytic spine

The Foundation and Fixed Points series give a dimension-neutral language for spectral selection, stable evolution, controlled disturbances, curved projectors, covariance, and exits from an admissible regime. Earlier confluences among stabilization time, physical time, and renormalization scale have been removed.

An exact local geometry

The world-in-world comparison field has a precise nine-component decomposition into rotation and strain. With a selected flag, the six strains split into one scalar, two diagonal shapes, and three shears. This is an exact mathematical result, although its physical identification remains conditional.

A selected global candidate

The q79 degree-three cover supplies an exact one-two-three trace carrier, finite arithmetic selection, local signed-sheet spin data, and a developed Fu–Yau/HYM setting. It is not merely the old Lens–Nil model under a new name.

A broad Standard Model reconstruction

The finite algebra, response matrix, flavor and mixing profiles, electroweak/Higgs/threshold rows, finite Dirac operator, multi-loop transport, and precision covariance form one audited construction. Embedded renormalized Standard Model equivalence is closed at the adopted one-shared-physical-primitive/profile tier.

A culture of correction

Several attractive claims have been rejected by the program's own verification: the extra-circle dimension count, a Lorentzian metric built from a positive Gram form, the old few-TeV chain, automatic Born weights, literal identity of Lens–Nil and q79 topology, and completed Gaussian quantum gravity. Retiring a failed claim narrows the theory toward what can actually survive.

Chapter 27

What Has Not Yet Been Secured

The remaining work is not one vague demand to “finish the theory.” It has specific layers.

Connect the local and global geometry

The world-in-world strain carrier and q79 trace carrier need one global map that preserves their frames, metrics, connections, response operators, and normalized overlaps. Without it, they are two promising constructions with a shared rank pattern.

Decide the global spin branch

The local double-cover relations are known. The remaining global loops must be lifted and their central signs checked. The result may be strict Spin, SpinC with shared-circle cancellation, or an obstruction to the intended branch.

Select one action and value source

One microscopic q79/Fu–Yau source should emit the normalized modes, couplings, thresholds, and response rows used downstream. It must do so without importing the values it is meant to explain.

Promote reconstruction to prediction

The current Standard Model result is broad profile equivalence. The upgrade is to freeze the source and predict held-out quantities. A theory with one to three genuinely selected primitives could still be a major reduction; zero knobs is an aspiration, not a rule imposed on nature.

Recover quantum probability and continuum dynamics

MTT needs a state and measurement framework whose probabilities equal the Born rule, together with a derivation or controlled recovery of the continuum quantum field theory and its gauge structure.

Derive gravity and cosmology from the same source

The selected action must produce the Lorentzian equations, constraints, stress response, Planck normalization, controlled four-dimensional reduction, cosmological solutions, perturbations, and observables.

Chapter 28

How the Program Can Fail

A useful fundamental proposal must expose itself to failure. MTT can fail in several concrete ways.

The q79 loops may obstruct the desired global spin structure. The world-in-world and q79 carriers may not admit a connection-preserving bridge. The selected action may fail anomaly cancellation or consistent reduction. The normalized overlap rows may miss the locked Standard Model profile. The apparent parameter reduction may disappear when every branch and normalization choice is counted. The proposed preprojection quantum language may fail to recover the Born rule or locality. The gravitational reduction may produce extra fields or incorrect observables.

There is also a broader danger: underdetermination. Many different upper constructions can be designed to reproduce the same lower data. Agreement then proves that a model is capable of representing nature, not that nature must use that model. MTT advances only when one source ties previously independent sectors together or predicts something held out.

This is why the current status language matters. A conditional theorem is a real theorem on its assumptions. A reconstruction is a real consistency achievement. A calibrated profile is a successful replay. A prediction is a different object. The categories are not a ranking of worth; they answer different questions.

In plain language. The theory succeeds only if common source relations replace independent inputs. A beautiful reinterpretation of known numbers is not enough.

Part VII

A Reader's Map

Chapter 29

Where to Find the Mathematics

This book intentionally leaves the proofs elsewhere. The following map points to the papers that own the main results.

Question	Technical owner
What is the abstract MTT architecture?	<i>Modal Triplet Theory: Foundations.</i>
When do projected states exist, persist, or become unique?	<i>Fixed Points I–VI</i> , with each paper retaining its standalone hypotheses and scope.
When does upper dynamics descend or permit recovery?	<i>The Projection–Admissibility Principle.</i>
What is known about four-plus-six signature compatibility?	<i>Lorentzian Base Compatibility and Signature Stability.</i>
What is the exact local 3-by-3 and one-two-three geometry?	<i>World-in-World Genesis</i> and <i>Closure-Strain Geometry.</i>
What is the proto-spinor result?	<i>The Proto-Spinor: Conditional Spinorial Closure and the q79 Interface.</i>
What is the selected global compactification candidate?	The q79, Fu–Yau, HYM, spin-lift, and foundational-geometry packets.
What action is currently available?	<i>Closure Geometry and a Regime-Local Ten-Dimensional Action Ansatz.</i>
What is the current Standard Model closure tier?	The consolidated True SM Closure ledger and its finite-operator, multi-loop, and precision certificates.
What happened to the few-TeV chain?	The corrected Theta Closure and Execution papers.
How are relationships to other theories typed?	<i>Modal Triplet Theory: A Typed Relationship Atlas.</i>

Chapter 30

A Small Status Guide

The reader can understand most of the current program with three labels.

Established within the stated model. A mathematical result has been proved or exactly verified once its assumptions and selected carrier are supplied. Examples include the local strain decomposition, several fixed-point and control results, finite algebra checks, and exact q79 branch packets.

Reconstructed at a declared tier. A familiar physical framework or data profile is reproduced after specified physical inputs are admitted. The current Standard Model result belongs here at the one-shared-physical-primitive/profile tier.

Interpretive or open. The mathematics suggests a physical picture, but the source map, dynamics, probability law, or observable test is incomplete. The deepest claims about particles, quantum ontology, gravity, time, and cosmology still contain work of this kind.

These labels keep the book readable without pretending that every sentence has the same logical status.

Chapter 31

Afterword: One Coherent World

The original book was driven by a simple dissatisfaction. Physics should not look like a cabinet of unrelated rules. Particles, forces, quantum states, spacetime, and measurement all belong to one world. A deeper theory should show why their descriptions fit together.

That dissatisfaction still drives MTT. What has changed is the understanding of how difficult the task is. A recurring number is not an explanation. A fixed point is not automatically an equilibrium. A phase circle is not time. A local double cover is not a global particle theory. A successful matrix reconstruction is not yet the unique source of its entries. A softened propagator is not quantum gravity.

These corrections do not empty the picture. They reveal its strongest form. MTT now has a reusable analytic spine, an exact local comparison geometry, a selected global q79/Fu–Yau candidate, a finite operator architecture, and a broad Standard Model reconstruction at a declared tier. More importantly, it has a short list of maps that would tie these achievements together.

The interpretive proposal can therefore be stated without spectacle:

Reality may admit a richer geometric and spectral description in which the stable theories we use are compatible projections. The recurring one-two-three structures may be stages of one shared carrier. Particles may be persistent modes, gauge forces rules of comparison, and gravity the visible response of the carrier to physical stress. The current mathematics supports parts of this picture and tells us exactly which bridges remain.

This book is an invitation to see that possibility. The proofs, failures, and decisions belong to the research papers. The next decisive advance will come when one of the missing bridges is constructed from the same source and survives an independent test.

Computational Evidence and Reproducibility

The Book is interpretive and proves no theorem. The closed, profile, and certified-numerical rows below are used directly only as examples of what the technical corpus currently contains. Their tier labels are preserved verbatim, and the open strict-upgrade ledger is presented as unfinished work rather than an achievement.

The referenced rows are frozen to the curated results repository state identified below. Hashes are grouped in eight-character blocks for line breaking.

Repository: <https://github.com/PeterNero/mtt-results-repro>

Commit: 31247ebb 5c22f3fb b5443024 365433c6 ee0bff4a

Manifest: release/result_manifest.json

Manifest SHA-256:

fb399689 60b00584 631dbf53 1a708e18

ef928d6b 6d935119 c185d7f6 32b1e7cd

Tier labels are quoted verbatim from that manifest. A row used directly supports only the specific computational statement identified above; a corpus-state cross-check does not prove this paper's local theorems; and an open row is evidence of an unresolved obligation, never of closure.

Rows used directly in this paper.

- A01/charged_yukawa_higgs_profile (*profile replay*).
Versioned Yu, Yd, Ye and lambda_H profile packet.
- A14/ckm_prediction_profile (*numeric certified*).
Three selected CKM profile rows and uncertainty comparison.
- A01/current_global_lock (*profile replay*).
Current non-looping global status and source-certificate map.
- A01/direct_k_higgs_row (*derived exact*).
Promoted direct K_threshold.Omega_H.lambda row.
- A22/e6_qpsi_qcd_anomaly (*derived exact*).
E6 Qpsi matter/exotic QCD anomaly cancellation audit.
- A04/final_12_of_12_audit (*profile replay*).
Twelve-obligation embedded renormalized-SM equivalence audit.
- A13/gr_tt_support (*derived exact*).
Exact-branch internal TT support certificate; physical normalization remains open.
- A19/hym_wiener_contraction (*numeric certified*).
Weighted-theta Fourier-tail and Wiener contraction certificate.

- A07/literal_cech_witness (*derived exact*).
Literal 81-entry, 729-cocycle finite Cech witness.
- A40/neutral_two_primitive_profile (*profile replay*).
Measured two-splitting neutrino profile, masses and Dirac Yukawa rows.
- A02/precision_15_source_transport (*profile replay*).
Fifteen measured source coordinates, Jacobian and covariance transport.
- A02/precision_8x8_workspace (*profile replay*).
Eight-coordinate SMDR output with positive-definite 8x8 covariance.
- A11/q79_exact_audit (*derived exact*).
Executable q=79 exact-branch audit.
- A11/q79_exact_theorem (*derived exact*).
CRT q=79 theorem on the selected exact branch.
- A01/qutrit_weyl_27_matrix (*derived exact*).
Sparse 27x27 qutrit-Weyl left-action realization.
- A01/strict_pew_row (*derived exact*).
Promoted P_EW source row at the declared one-shared-primitive standard.

Open boundary (not evidence of closure).

- A05/strict_upgrade_ledger (*open*).
Current 2/9 strict no-knob upgrade ledger.

No imported row changes theorem ownership or promotes a neighboring claim: all local statements retain their stated hypotheses, domains, and limitations.

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